



國立清華大學
NATIONAL TSING HUA UNIVERSITY

Deep Learning and Applications

Ching-I Huang

Maximum a Posterior (MAP)

- Problem with MLE:
 1. Relies only on observed data
 2. Can lead to **overfitting**, especially when data is small or noise.
- Purpose of maximum a posterior: Incorporate prior knowledge about parameters $\mathbf{w}$.

$$\hat{\mathbf{w}}_{MAP} = \arg \max_{\mathbf{w}} p(\mathbf{w} | \mathcal{D}) \quad (\text{given } \mathcal{D} \text{ find } \mathbf{w})$$

Combine data (likelihood) with prior belief to estimate parameters.

- Assuming i.i.d. data drawn from a Gaussian distribution

$$p(y | x) = \prod_i^m G(y; \mu, \sigma^2)$$

Product rule of probability

$$P(a, b, c) = P(a|b, c)P(b|c)P(c)$$

Conditional density function

$$P(\mathbf{x} = x | \mathbf{y} = y) = \frac{P(\mathbf{x} = x, \mathbf{y} = y)}{P(\mathbf{y} = y)}, \text{ only when } P(\mathbf{y} = y) > 0.$$

- **Probability mass function:**

If $\mathbf{x}$ is discrete, probability mass function $P(\mathbf{x}=x)$ or $P(x)$, ex. the output of a fair dice has discrete uniform distribution with $P(x) = 1/6$.

- **Probability density function:**

If $\mathbf{x}$ is continuous, $P(\mathbf{x} = x)$ denotes a probability density function $p(x)$, rate of increase in probability at x .

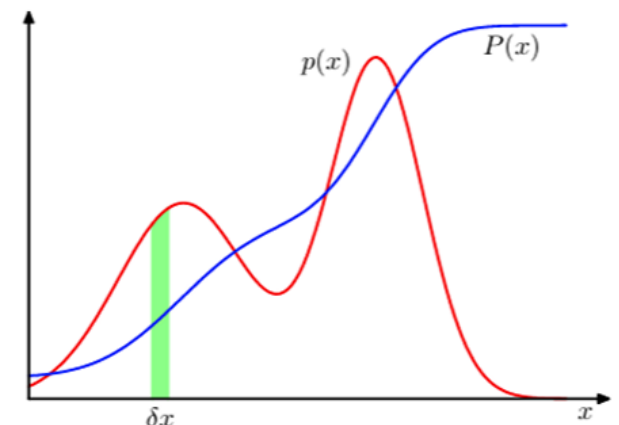


Fig. 1: The probability density $p(x)$ can be expressed as the derivative of a cumulative distribution function $P(x)$.



Maximum a Posterior (MAP)

- Bayes' Theorem

$P(\mathbf{y} \mathbf{x}) = \frac{P(\mathbf{x} \mathbf{y})P(\mathbf{y})}{P(\mathbf{x})}$	$P(\mathbf{x}, \mathbf{y}) = P(\mathbf{y} \mathbf{x}) P(\mathbf{x}) = P(\mathbf{x} \mathbf{y}) P(\mathbf{y})$
$posterior = \frac{likelihood \times prior}{evidence}$	e.g.: $\mathbf{x}$: symptoms; $\mathbf{y}$: disease; $P(\mathbf{y})$: the probability of getting this disease $P(\mathbf{x} \mathbf{y})$: probability of this disease having this symptom.

- Posterior

Function with variable $\mathbf{w}$

$$p(\mathbf{w} | \mathcal{D}) = \frac{\text{Likelihood } p(\mathcal{D} | \mathbf{w}) \times \text{Prior } p(\mathbf{w})}{\text{Evidence } p(\mathcal{D})}$$

Posterior probability $p(\mathbf{w} | \mathcal{D})$: updated belief after seeing data

Likelihood function $p(\mathcal{D} | \mathbf{w})$: fit to data (related to some known parameters based on the observed data)

Prior probability $p(\mathbf{w})$: The probability of $\mathbf{w}$ available before observation of the events.

Evidence $p(\mathcal{D})$: normalization constant (independent of $\mathbf{w}$)

Integrate w.r.t $\mathbf{w} \rightarrow \int p(\mathbf{w}|D)d\mathbf{w} = \int \frac{p(D|\mathbf{w})p(\mathbf{w})}{p(D)} d\mathbf{w} \rightarrow p(D) = \int p(D|\mathbf{w})p(\mathbf{w})d\mathbf{w}$

Total area of posterior probability distribution = 1
 $p(D)$ is constant (Not related to $\mathbf{w}$)

Posterior probability $p(\mathbf{w}|D) = \frac{p(D|\mathbf{w})p(\mathbf{w})}{\int p(D|\mathbf{w})p(\mathbf{w})d\mathbf{w}} \rightarrow \text{Posterior} \propto \text{Likelihood} \times \text{Prior}$



Maximum a Posterior (MAP)

- Maximize the possibility of the posterior function to get the most possible parameters $\mathbf{w}$ with the **consideration of prior probability**

$$\arg \max_{\mathbf{w}} P(\mathbf{w}|\mathbf{X}) = \arg \max_{\mathbf{w}} \frac{P(\mathbf{X}|\mathbf{w})P(\mathbf{w})}{P(\mathbf{X})} \quad \text{where } P(\mathbf{w}) \text{ is prior knowledge about } \mathbf{w}.$$

$$= \arg \max_{\mathbf{w}} P(\mathbf{X}|\mathbf{w})P(\mathbf{w})$$

$$= \arg \max_{\mathbf{w}} \log P(\mathbf{X}|\mathbf{w})P(\mathbf{w})$$

$$= \arg \max_{\mathbf{w}} \log P(\mathbf{X}|\mathbf{w}) + \log P(\mathbf{w})$$

Assume $P(\mathbf{w}) \sim G(0, \alpha^{-1}\mathbf{I})$

$$P(\mathbf{w}) = \left(\frac{\alpha}{2\pi}\right)^{\frac{M}{2}} \exp\left(-\frac{\alpha}{2}\mathbf{w}^T\mathbf{w}\right)$$

Then,

$$\arg \max_{\mathbf{w}} P(\mathbf{w}|\mathbf{X}) =$$

$$= \arg \min_{\mathbf{w}} \frac{\beta}{2} \sum_{n=1}^N (y_n - \mathbf{w}^T \mathbf{x}_n)^2 + \frac{\alpha}{2} \mathbf{w}^T \mathbf{w}$$

$$= \arg \min_{\mathbf{w}} \sum_{n=1}^N (y_n - \mathbf{w}^T \mathbf{x}_n)^2 + \frac{\lambda}{2} \mathbf{w}^T \mathbf{w}$$

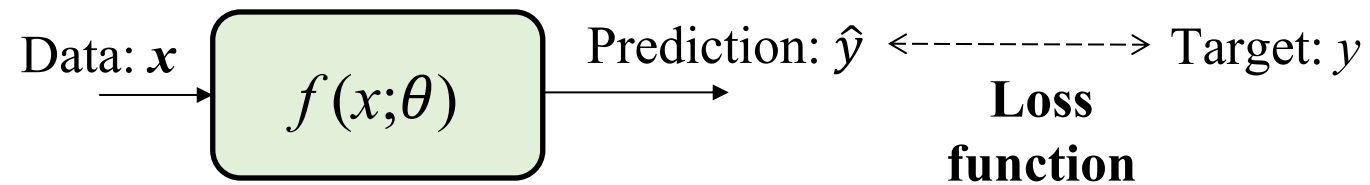
If $p(w)$ is uniform (constant), MAP reduces to MLE.

If $p(w)$ is Gaussian, MAP corresponds to MLE with L2 regularization (weight decay).



Supervised Learning Tasks

- Supervised Learning Tasks: The performance at task improves with experience.



- Regression: predict a numerical value given some input

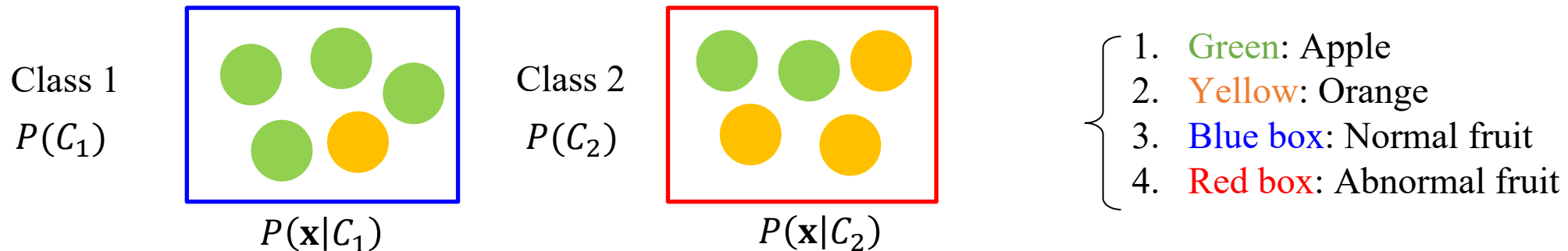
$$f : \mathbb{R}^n \rightarrow \mathbb{R}^m \text{ e.g., temperature forecasting}$$

- Classification: predict a categorical label given some input

$$f : \mathbb{R}^n \rightarrow \{1, \dots, k\} \text{ e.g., classifying machine parts as defective or non-defective}$$

Classification (1.2 & 4.2)

- The task of assigning an input x to one of a finite set of categories $y \in \{1, 2, \dots, K\}$.
- Binary Classification** $K = 2$, i.e., $y \in \{0, 1\}$.
- Given a fruit x , which box (Class C_1 , C_2) does it belong to?



- Generative model:** Probability that a given x belongs to Class 1 (Normal fruit)

$$P(C_1 | \mathbf{x}) = \frac{P(\mathbf{x} | C_1) P(C_1)}{P(\mathbf{x} | C_1) P(C_1) + P(\mathbf{x} | C_2) P(C_2)} = P(\mathbf{x}) = \frac{P(\mathbf{x} | C_1) P(C_1)}{P(\mathbf{x})}$$

Self-generate sample x with the probability

$$P(\mathbf{x} | C_1) \quad P(\mathbf{x} | C_2)$$

If a sample truly comes from C_1 or C_2 what is the probability distribution of its features $\mathbf{x}$

$$\Rightarrow P(\mathbf{x} | C_1) = \mathcal{N}(\mathbf{x} | \mu_{C_1}, \Sigma_{C_1})$$

$$P(\mathbf{x} | C_2) = \mathcal{N}(\mathbf{x} | \mu_{C_2}, \Sigma_{C_2})$$

Assume each class follows a distribution and estimate $P(\mathbf{x} | C_1)$, $P(\mathbf{x} | C_2)$ using MLE (e.g., Gaussian).



With $P(\mathbf{x}|C_1)$, $P(\mathbf{x}|C_2)$, and empirical prior $P(C_1) = N_1/N$, $P(C_2) = N_2/N$, the classification can be performed !

N : Total number of fruit



Classification

- Posterior probability with Gaussian assumption

$$P(C_1 | \mathbf{x}) = \frac{P(\mathbf{x} | C_1)P(C_1)}{P(\mathbf{x} | C_1)P(C_1) + P(\mathbf{x} | C_2)P(C_2)} = \frac{1}{1 + \frac{P(\mathbf{x} | C_2)P(C_2)}{P(\mathbf{x} | C_1)P(C_1)}} = \frac{1}{1 + e^{-z}} = \sigma(z)$$

Using sigmoid is not a heuristic—it is the result of sound probabilistic modeling using Bernoulli likelihood.

where the log-odds z is defined as

$$z = \ln \frac{P(\mathbf{x} | C_1)P(C_1)}{P(\mathbf{x} | C_2)P(C_2)} = \ln \frac{P(\mathbf{x} | C_1)}{P(\mathbf{x} | C_2)} + \ln \frac{P(C_1)}{P(C_2)}.$$

Assume the class-conditional distributions are Gaussian with the same covariance matrix Σ :

$$P(\mathbf{x} | C_k) = \mathcal{N}(\mathbf{x} | \boldsymbol{\mu}_k, \Sigma), \quad k = 1, 2, \quad G(\mathbf{x} | \boldsymbol{\mu}, \Sigma) = \frac{1}{(2\pi)^{M/2}} \frac{1}{|\Sigma|^{1/2}} \exp \left\{ -\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu})^T \Sigma^{-1} (\mathbf{x} - \boldsymbol{\mu}) \right\}$$

Then the quadratic terms $\frac{1}{2} \mathbf{x}^T \Sigma^{-1} \mathbf{x}$ cancel, and the log-odds become a linear function of $\mathbf{x}$:

$$z = (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)^T \Sigma^{-1} \mathbf{x} - \frac{1}{2} \boldsymbol{\mu}_1^T \Sigma^{-1} \boldsymbol{\mu}_1 + \frac{1}{2} \boldsymbol{\mu}_2^T \Sigma^{-1} \boldsymbol{\mu}_2 + \ln \frac{N_1}{N_2}.$$

Define

$$\mathbf{w} = \Sigma^{-1}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2), \quad b = -\frac{1}{2} \boldsymbol{\mu}_1^T \Sigma^{-1} \boldsymbol{\mu}_1 + \frac{1}{2} \boldsymbol{\mu}_2^T \Sigma^{-1} \boldsymbol{\mu}_2 + \ln \frac{N_1}{N_2}$$

so that the posterior probability can be written as

$$P(C_1 | \mathbf{x}) = \sigma(\mathbf{w}^T \mathbf{x} + b)$$

Linear

Under the Gaussian assumption with shared covariance, the posterior probability of class C_1 is given by the **sigmoid of a linear function of the input features**.

Requires fewer parameters, thereby reducing the risk of overfitting.



Binary Classification with Logistic Regression

- Posterior probability is modeled as

$$P(C_1 | \mathbf{x}; \mathbf{w}, b) = P_{w,b}(C_1 | \mathbf{x}_n) = \sigma(\mathbf{w}^T \mathbf{x} + b)$$

Find the maximum posterior probability that fits the training data

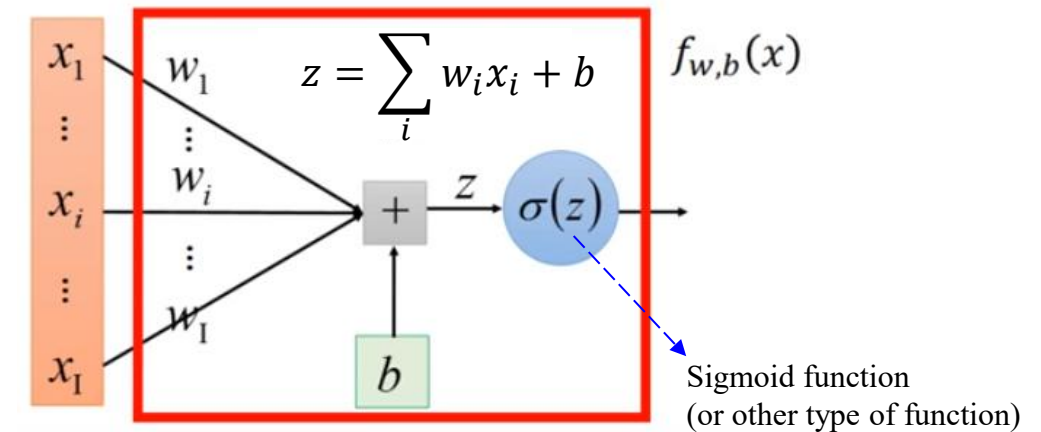
- Likelihood of data**

$$\begin{aligned} L_{w,b}(C_1 | \mathbf{x}_n) &= \prod_{n=1}^N \left[\frac{P_{w,b}(C_1 | \mathbf{x}_n)^{t_n}}{y_n} \frac{(1 - P_{w,b}(C_1 | \mathbf{x}_n))^{1-t_n}}{1 - y_n} \right] \\ &= \prod_{n=1}^N y_n^{t_n} (1 - y_n)^{1-t_n}, \quad y_n = \sigma(\mathbf{w}^T \mathbf{x}_n + b) \end{aligned}$$

- Training objective**

$$\mathcal{L} = - \sum_{n=1}^N [t_n \ln y_n + (1 - t_n) \ln(1 - y_n)]$$

Minimize the negative log-likelihood (cross-entropy loss)



Training data

$\mathbf{x}_1$	$\mathbf{x}_2$	$\mathbf{x}_3$	$\dots$	$\mathbf{x}_n$
C_1	C_1	C_2	$\dots$	
$t_1=1$	$t_2=1$	$t_3=0$	$\dots$	t_n

$t_n = 1$ if $\mathbf{x}_n \in C_1$
 $t_n = 0$ if $\mathbf{x}_n \in C_2$